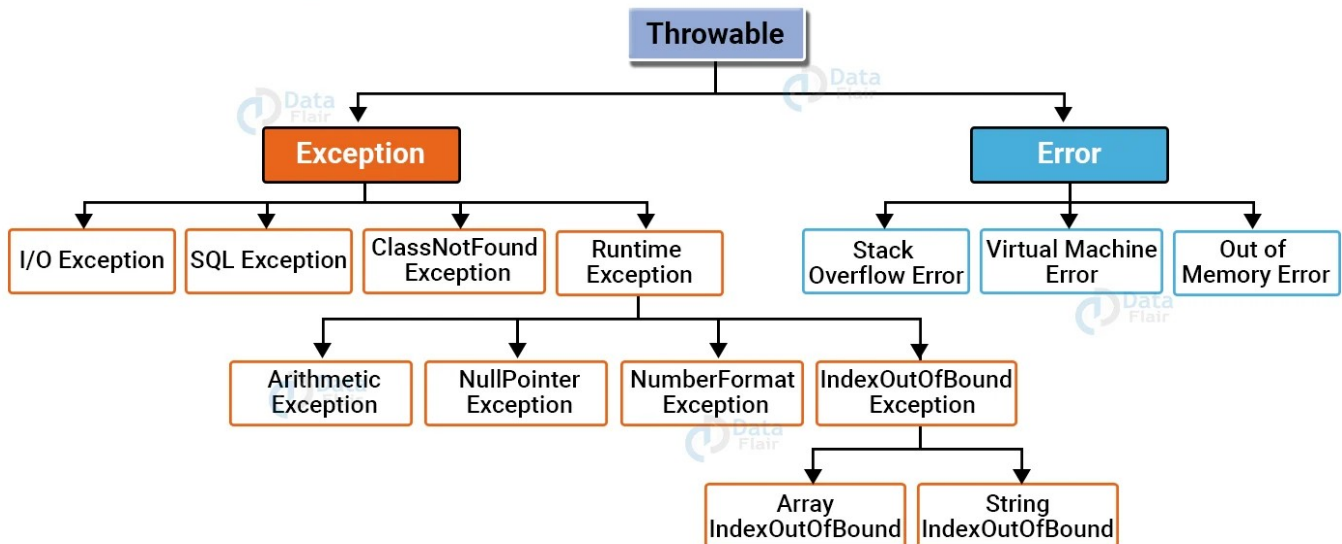


Exception Handling

1. What is Exception Handling in Java?

- ➔ Exception means Abnormal Condition.
- ➔ Exception Handling in Java is a powerful mechanism to handle abnormal condition which will occurs while running Java Program so that the normal flow of the application can maintained.
- ➔ Exception Handling is a mechanism to handle runtime errors such as ClassNotFoundException, IOException, SQLException, RemoteException, etc.

Hierarchy of Java Exceptions



2. What is explicit and implicit in java?

- ➔ **Explicit** means something is done by the programmer **Implicit** means that it's done by the compiler.

3. What is the difference between Checked and Unchecked Exception?

- ➔ Checked Exception occurs while Compiling Time like I/OException, SQLException, ClassNotFoundException.
- ➔ Unchecked Exception occurs Runtime like NullPointerException, NumberFormatException.

4. What is Try Catch Block in Java?

- ➔ Try-Catch Block is use for Handling Exception
- ➔ Try block is use for enclose the data that might be throw an Exception.
- ➔ Catch block is use to catch those exception thrown by try block.
- ➔ Try-Catch Block is must be in within the Method.
- ➔ Try block must having catch or finally Block.

The JVM firstly checks whether the exception is handled or not. If exception is not handled, JVM provides a default exception handler that performs the following tasks:

- ➔ Prints out exception description.
- ➔ Prints the stack trace (Hierarchy of methods where the exception occurred).
- ➔ Causes the program to terminate.

5. Can we Use Multiple Catch Block and Why?

- ➔ Yes we can Use multiple Catch Block.
- ➔ Each Block contain Different Exception Handler.
- ➔ If there is possibility to have multiple exception will occurs then we will use multiple Catch Block.

6. What is Nested Try-Catch Block in Java and why we use and how it will Work?

- ➔ As name suggest Nested Try-Catch Block mean's Try-Catch block in Try Block.
- ➔ If we want use particular Exception handle in some block of code in Try Block then we use Nested try-catch.
- ➔ If inner try catch block dose not handle exception the outer will handle it.
- ➔ If any inner catch block will not handle then main try catch block will handle.
- ➔ If none of the catch block will unable to handle then Java runtime Exception handle the Exception.

7. What is Finally in Java?

- ➔ Finally is a block which is use to execute important code like closing of connection.
- ➔ Finally block is always execute whether exception is handle or not.

8. Is there any possibility when finally block is not executed?

- ➔ Yes, If before calling finally block there is System.exit(0); Statement is present.

9. Difference between Final, Finally and Finalize

Final	Finally	Finalize
Final keyword is use to restrict the class, method, or variables.	Finally is a block which will execute whether the exception occurs or not.	Finalize is the method of Java which will perform to clean up process just before Garbage collected.
Final method will execute when we call.	Finally block will always execute if before this System.exit() is not present.	Finalize method is execute just before the object is destroyed.
(1) Once declared, final variable becomes constant and cannot be modified. (2) final method cannot be overridden by sub class. (3) final class cannot be inherited.	(1) finally block runs the important code even if exception occurs or not. (2) finally block cleans up all the resources used in try block	finalize method performs the cleaning activities with respect to the object before its destruction.
final int age = 18;	// executes regardless of exception occurred or not finally { System.out.println("finally block is always executed"); }	// defining the finalize method protected void finalize() { System.out.println("Called the finalize() method"); }

10. What is the difference between the throw and throws keyword?

Throw	Throws
Throw is keyword used to explicitly throw the exception within a method or block of code.	Throws is used in Method signature to declare the type of exception that a method may throw
<pre>public void someMethod() { if (condition) { throw new SomeException("Some error occurred"); } }</pre>	<pre>public void someMethod() throws SomeException { // method implementation }</pre>

11. What is Java Custom Exception and How and Why we use?

- We can create our own Exception that is derived class of the Exception class is called as Custom Exception.
- We can create this for to catch and provide specific treatment for application.
- Also to understand if any specific exception will happen in Business Logic so we can catch this exact problem in application and help to understand a developer.

```
public class WrongFileNameException extends Exception {  
    public WrongFileNameException(String errorMessage) {  
        super(errorMessage);  
    }  
}
```

Exception handling with Method Overriding

- If Superclass Method dose not declare an Exception then Subclass overridden method cannot declare the checked exception but it can declare unchecked exception.
- If superclass method declare an exception the subclass overridden method can declare same exception, subclass exception or no exception but cannot declare parent exception.